

The Multivariate Normal Distribution

2026-04-17

Introduction

The multivariate normal (MVN) is the joint distribution that plays the same role for random vectors that the normal plays for random scalars. It is parameterised by a mean vector and a covariance matrix, and it underlies linear discriminant analysis, principal component analysis, Gaussian graphical models, and most of multivariate inference.

Prerequisites

Normal distribution, linear algebra (vectors, matrices, positive-definite matrices).

Theory

A random vector $\mathbf{X} \in \mathbb{R}^p$ is MVN with mean $\boldsymbol{\mu} \in \mathbb{R}^p$ and positive definite covariance $\Sigma \in \mathbb{R}^{p \times p}$ if its density is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right\}.$$

The exponent is the squared **Mahalanobis distance** from $\mathbf{x}$ to $\boldsymbol{\mu}$.

Key properties:

- **Marginals are MVN:** every subset of components is MVN.
- **Conditionals are MVN:** given a subset of components, the rest is MVN with mean and covariance depending on the conditioning values.
- **Linear transformations:** if $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$ and A is any matrix, then $A\mathbf{X} \sim \mathcal{N}(A\boldsymbol{\mu}, A\Sigma A^\top)$.
- **Zero correlation implies independence** in the MVN family (unlike general distributions).
- **Mahalanobis distance squared** $\sim \chi_p^2$ for $\mathbf{X} \sim \mathcal{N}_p(\boldsymbol{\mu}, \Sigma)$.

Assumptions

Σ must be symmetric positive-definite. Degenerate cases (singular Σ) give a distribution supported on a lower-dimensional affine subspace.

R Implementation

```
library(mvtnorm); library(ggplot2)

set.seed(2026)
mu      <- c(0, 1)
Sigma   <- matrix(c(1, 0.6, 0.6, 2), 2, 2)

X <- rmvnorm(2000, mean = mu, sigma = Sigma)
colMeans(X)
cov(X)
```

```
df <- data.frame(X1 = X[, 1], X2 = X[, 2])
ggplot(df, aes(X1, X2)) +
  geom_point(alpha = 0.3, colour = "#2A9D8F") +
  stat_ellipse(level = 0.95, colour = "#F4A261", linewidth = 1) +
  coord_equal() +
  theme_minimal()

# Conditional distribution:  $X_1 \mid X_2 = x_2$ 
#  $\mu_{1|2} + \Sigma_{12} \Sigma_{22}^{-1} (x_2 - \mu_2)$ 
x2_val <- 2.0
cond_mean <- mu[1] + Sigma[1, 2] / Sigma[2, 2] * (x2_val - mu[2])
cond_var <- Sigma[1, 1] - Sigma[1, 2]^2 / Sigma[2, 2]
c(mean = cond_mean, var = cond_var)

# Mahalanobis distance
md2 <- mahalanobis(X, center = mu, cov = Sigma)
c(mean_md2 = mean(md2), expected = 2) #  $\chi^2_2$  mean = 2
```

Output & Results

```
[1] 0.003 1.012

      [,1]      [,2]
[1,] 1.002 0.605
[2,] 0.605 1.989

      mean      var
0.3000 0.8200

mean_md2 expected
2.00      2.00
```

Empirical mean and covariance closely match the theoretical values, and the Mahalanobis distance squared has the expected chi-squared mean.

Interpretation

MVN assumptions underlie multiple regression (normal errors), linear discriminant analysis (normal class densities), and most factor-analytic and SEM approaches. Violations of joint normality invalidate ellipsoidal confidence regions and some likelihood-based tests.

Practical Tips

- Check joint normality with chi-squared Q-Q plots of Mahalanobis distance squared, not just marginal Q-Q plots.
- Near-singular Σ (high collinearity) causes numerical instability; regularise with $\Sigma + k \mathbf{I}$ or use generalised inverses.
- For high-dimensional MVN (p large), shrinkage estimators (Ledoit-Wolf) stabilise the covariance.
- Sampling from MVN: `mvtnorm::rmvnorm`, or directly via Cholesky: $\mathbf{X} = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}$ with $\mathbf{L}\mathbf{L}^\top = \Sigma$ and $\mathbf{z} \sim \mathcal{N}_p(0, \mathbf{I})$.
- The Kullback-Leibler divergence between two MVNs has a closed form; useful in information-geometric methods.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots